

MedInsure Case Study: Ensuring Model Reproducibility

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MedInsure Case Study – Ensuring Model Reproducibility (Q5)

Data Versioning, Environment Metadata, Hyperparameter Tracking



ML Integrity

Maintaining the reliability and consistency of machine learning pipelines.



Healthcare Compliance

Adhering to strict regulatory requirements in the healthcare sector.



Audit Readiness

Preparing for rigorous external reviews and validation processes.

What is Model Reproducibility?

Model reproducibility is the ability to recreate the exact same machine learning model, achieving identical results using the same data, computational environment, and hyperparameters. It's a cornerstone for trust and reliability in AI systems.

→ **Compliance & Transparency**

Ensuring models meet regulatory standards and can be transparently validated.

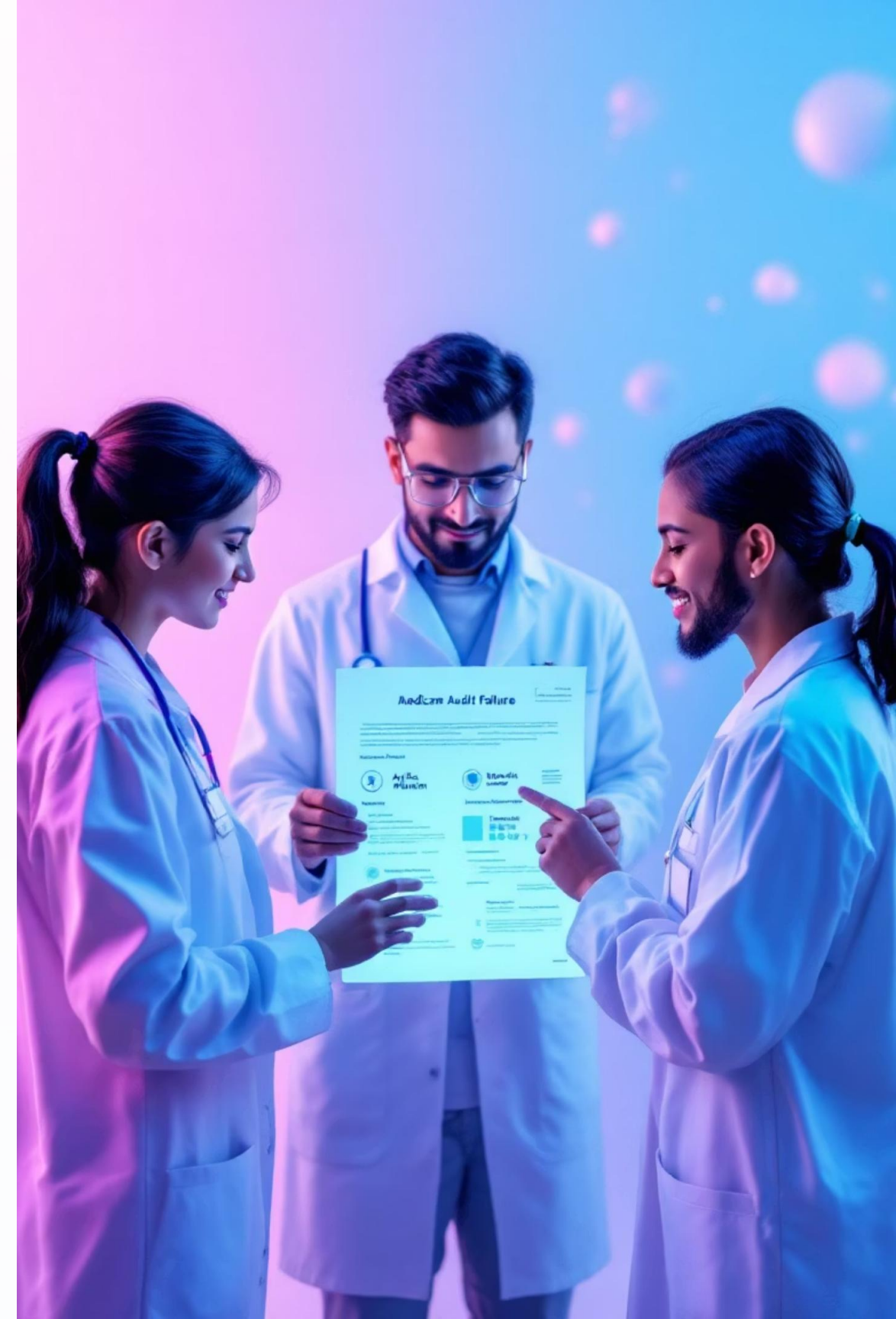
→ **Debugging & Iteration**

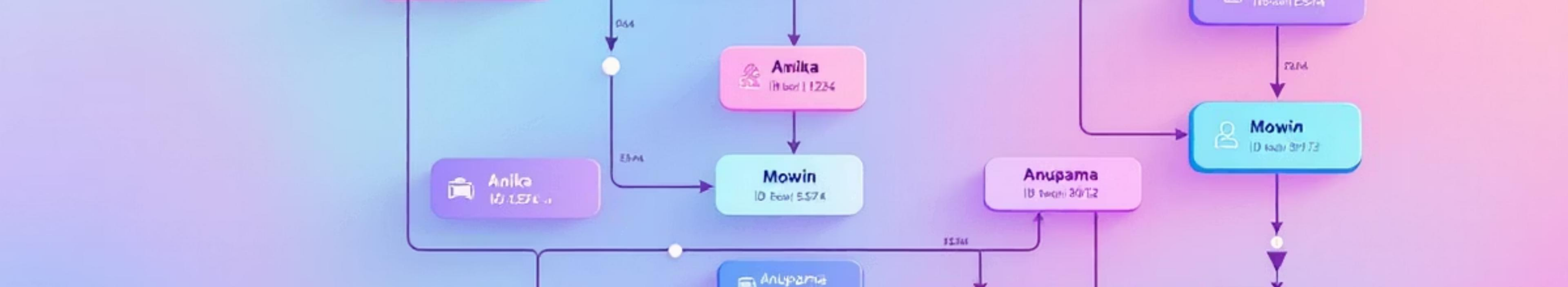
Facilitating the identification and resolution of issues, speeding up development cycles.

→ **Stakeholder Trust**

Building confidence among users and regulators in the model's reliability and fairness.

During a recent audit, MedInsure encountered significant challenges in reproducing its claim-approval model, highlighting critical gaps in its MLOps practices.





Audit Scenario: A Reproducibility Breakdown

During a routine audit, MedInsure faced a critical test: retraining an old claim-approval model. The results exposed deep-seated issues in their MLOps processes.

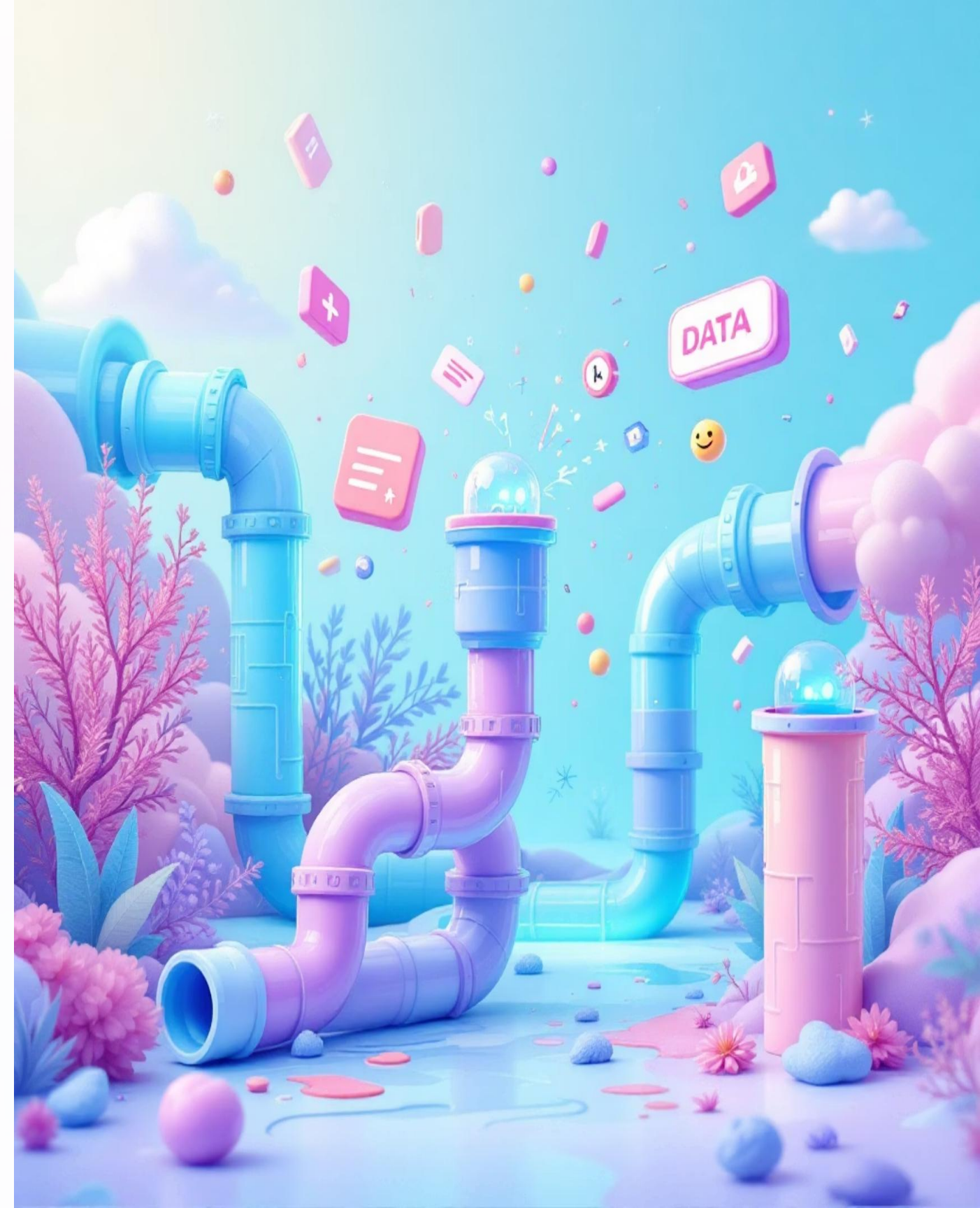
01	02	03
Auditor's Request	Reproduction Attempt Fails	Root Causes Identified
An external auditor requested MedInsure to retrain an existing claim-approval model to validate its performance and logic against current data.	The data science team attempted to reproduce the model, but despite using the "same" code, the retrained model yielded significantly different and inconsistent results.	Investigation revealed that missing historical logs and mismatched software environments were the primary culprits, preventing accurate reproduction.

Issue 1: Missing Training Data Versions

A major roadblock in MedInsure's reproducibility efforts was the inability to access the exact training dataset used for the original model development.

- The specific dataset utilized months prior for initial model training was no longer readily available in its original form.
- Unrecorded ETL (Extract, Transform, Load) changes inadvertently removed a crucial feature, "hospital_risk_score," altering the dataset's composition.
- Without a robust versioning system, the team found it impossible to recover the precise historical dataset, leading to irreproducible training.

The impact was significant: without the original data, the model could not be recreated as it was, invalidating any attempts at true reproducibility.



Fix 1: Implementing Robust Data Versioning

To prevent future data-related reproducibility failures, MedInsure must adopt dedicated data versioning tools and practices.



Tools for Versioning

Utilize systems like DVC (Data Version Control) or LakeFS to track and manage changes in datasets, treating them like code.



Comprehensive Metadata

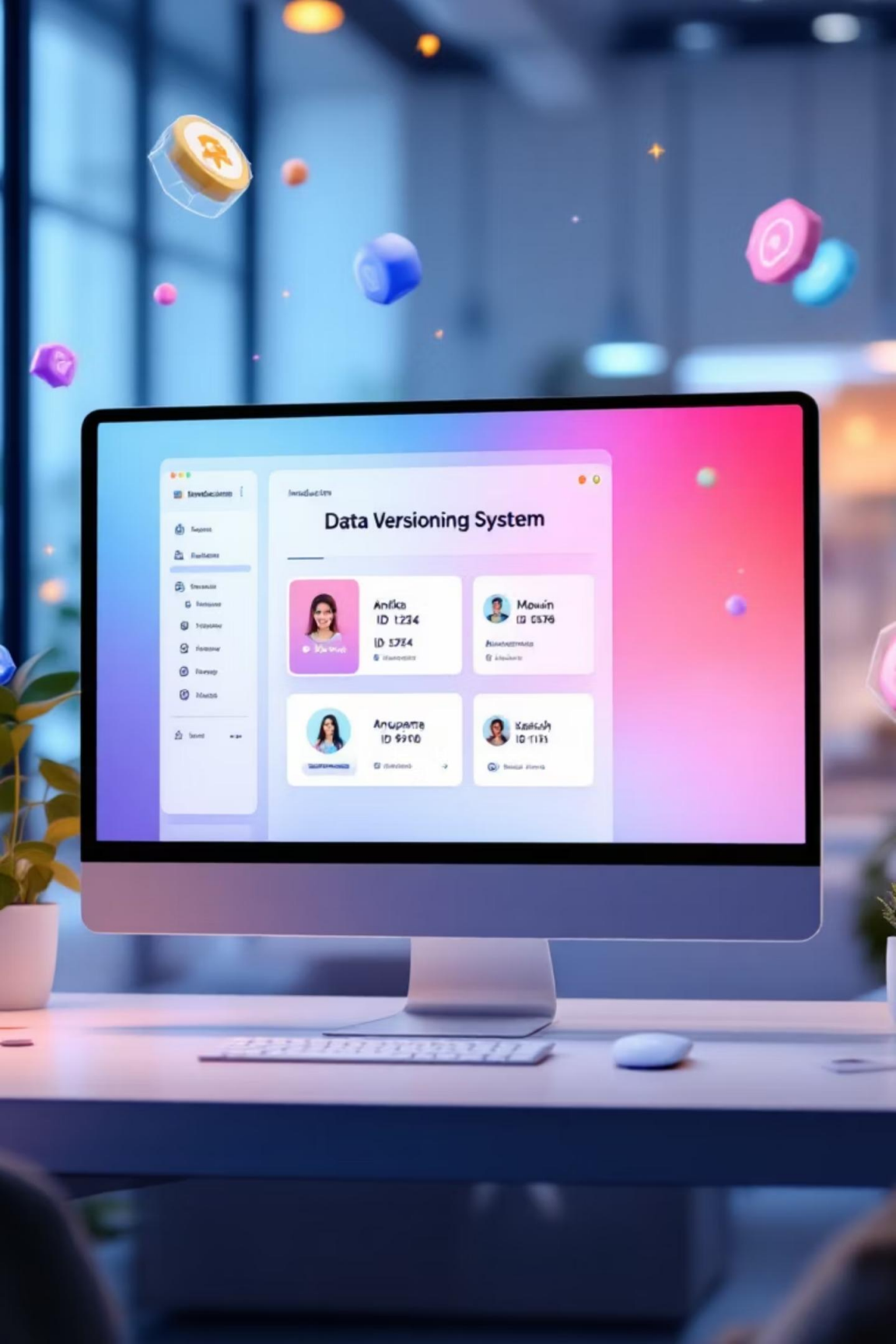
Store detailed metadata including dataset IDs, timestamps, and the exact preprocessing scripts used for each data version.



Immutable Data Copies

Maintain an immutable, timestamped copy of every dataset used for model training, ensuring historical data integrity.

Outcome: With data versioning, the original dataset can be restored anytime, guaranteeing the foundation for model reproducibility and audit readiness.



Issue 2: Environment Mismatch Leading to Inconsistent Predictions

Discrepancies in the software environment proved to be another critical barrier to reproducing MedInsure's ML model, causing significant operational challenges.

Training Environment: The original model was trained using Python 3.8 and scikit-learn version 0.24.

Production Environment: The retraining attempt was conducted in a different environment, featuring Python 3.10 and scikit-learn version 1.2

Result: Even with the "same" data and code, the model's predictions diverged substantially due to library version differences and Python interpreter updates.



Fixes: Environment & Hyperparameter Tracking

MedInsure can achieve full reproducibility by standardizing environments and diligently tracking every aspect of model development.

Environment Fixes:

Containerization with Docker: Encapsulate the entire training environment within Docker containers to ensure consistency across all stages.

Dependency Management: Meticulously maintain a `requirements.txt` file and capture full OS metadata for every environment build.

Random Seed Management: Always store and utilize random seeds to ensure consistent initialization across experiments.

Hyperparameter Fixes:

Experiment Tracking Platforms: Implement tools like MLflow or Weights & Biases to log all hyperparameters, training configurations, and model artifacts.

Traceable Model Versions: Ensure every iteration and version of the model, along with its associated metadata, is meticulously tracked and easily retrievable.



Summary: The Path to Reproducible ML for MedInsure

Ensuring full ML model reproducibility is paramount for MedInsure, addressing compliance, operational efficiency, and stakeholder trust.



These practices ensure:

Audit Readiness: Seamlessly meet regulatory and internal auditing requirements.

Consistent Predictions: Deploy models with predictable and reliable outcomes.

Model Transparency: Foster a clear understanding of model behavior and decisions.

MedInsure must implement these strategies diligently to prevent future failures and build a robust, trustworthy AI framework.